

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/NTR

Original Research

Evaluating reliability, quality, and readability of ChatGPT's nutritional recommendations for women with polycystic ovary syndrome

Elif Ulug^{a,b}, Irmak Gunesli^c, Aylin Acikgoz Pinar^a, Bulent Okan Yildiz^{c,d,*}^aDepartment of Nutrition and Dietetics, Faculty of Health Sciences, Hacettepe University, 06100, Ankara, Turkey^bDepartment of Nutrition and Dietetics, Faculty of Health Sciences, Ataturk University, 25240, Erzurum, Turkey^cDepartment of Internal Medicine, Hacettepe University School of Medicine, 06100, Ankara, Turkey^dDivision of Endocrinology and Metabolism, Hacettepe University School of Medicine, 06100, Ankara, Turkey

ARTICLE INFO

Article history:

Received 7 August 2024

Revised 13 November 2024

Accepted 13 November 2024

Keywords:

Polycystic ovary syndrome

Nutrition

Diet

ChatGPT

Reliability

Quality

Readability

ABSTRACT

Patients with polycystic ovary syndrome (PCOS) often have many questions about nutrition and turn to chatbots such as Chat Generative Pretrained Transformer (ChatGPT) for advice. This study aims to evaluate the reliability, quality, and readability of ChatGPT's responses to nutrition-related questions asked by women with PCOS. Frequently asked nutrition-related questions from women with PCOS were reviewed in both Turkish and English. The reliability and quality of the answers were independently evaluated by 2 authors and a panel of 10 expert dietitians, using modified DISCERN and global quality score. Additionally, the readability of the answers was calculated using frequently used readability formulas. The mean modified DISCERN scores for English and Turkish versions were 27.6 ± 0.87 and 27.2 ± 0.87 , respectively, indicating a fair level of reliability in the responses (16–31 points or 40%–79%). According to the global quality score, 100% of the responses in English and 90.9% of the responses in Turkish were rated as high quality. The readability of responses was classified as “difficult to read” with the readership levels assessed at college level and above for both English and Turkish. The correlation and regression analyses indicated no relationship between reliability, quality, and readability in English. However, a significant relationship was observed between quality and readability indexes in Turkish ($P < .05$). Our results suggest that ChatGPT's responses to nutrition-related questions about PCOS are generally of high quality, but improvements in both reliability and readability are still necessary. Although ChatGPT can offer general information and guidance on nutrition for PCOS, it should not be

Abbreviations: AI, Artificial intelligence; AMA, American Medical Association; ChatGPT, Chat Generative Pretrained Transformer; FRE, Flesch Reading Ease; GQS, global quality score; mDISCERN, modified DISCERN; NIH, National Institutes of Health; PCOS, polycystic ovary syndrome.

* Corresponding author at: Hacettepe University School of Medicine, Department of Internal Medicine, Division of Endocrinology and Metabolism, Hacettepe 06100, Ankara, Turkey.

E-mail address: yildizbo@yahoo.com (B.O. Yildiz).

<https://doi.org/10.1016/j.nutres.2024.11.005>

0271-5317/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

considered a substitute for personalized medical advice from health care professionals for effective management of the syndrome.

© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

Polycystic ovary syndrome (PCOS) is a common endocrine and metabolic disorder characterized by androgen excess, ovulatory dysfunction, and polycystic ovarian morphology, affecting up to 1 in 5 premenopausal women [1]. Managing nutrition is one of the key strategies for preventing and treating metabolic disorders associated with PCOS [2], leading PCOS patients to have questions about nutrition [3]. Although many people looking for nutrition advice turn to the internet and other online platforms, artificial intelligence (AI) technology is rapidly emerging as a powerful alternative tool [4,5].

Technological advancements in AI have led to the development of chatbots that use large language models [6,7]. Chat Generative Pretrained Transformer (ChatGPT), a chatbot developed by OpenAI (San Francisco, CA, USA) in November 2022, is one of the most popular and widely recognized models. ChatGPT is continuously refined through training using the Reinforcement Learning from Human Feedback model [8]. In the modern world, where computer technologies are rapidly advancing, most people acquire information about their health conditions and possible treatment options from online platforms [9,10]. Chatbots such as ChatGPT have become increasingly preferred because they are available around the clock, easy to access and can provide instant answers [11]. AI systems can provide dietary recommendations [12]; therefore, expert evaluation of information provided by AI is essential.

Recent studies have evaluated the accuracy of ChatGPT's responses to patient inquiries regarding PCOS [13] and its related infertility [14]. Additionally, AI, including ChatGPT, has the potential to serve as a valuable tool for disseminating nutritional information [15]. Studies conducted recently suggest that ChatGPT offers reasonably accurate information regarding general dietary guidelines for a number of noncommunicable disorders [16–19]. Taking all of this into account, it is crucial to evaluate the reliability and accuracy of responses provided by well-known platforms such as ChatGPT, particularly given the frequent reliance of PCOS patients on online resources for nutritional information. Therefore, this study was conducted in response to a large gap in the current literature, with the goal of assessing the reliability, quality, and readability of ChatGPT's answers to nutrition-related questions from PCOS patients in both English and Turkish.

2. Methods

2.1. Study design

Research team collected the frequently asked questions of PCOS patients during previous studies [20,21]. A total of 63

questions were obtained, and the questions with similar content but expressed differently were combined to form 11 questions under 6 main topics. These main topics include general nutritional recommendations, weight loss/management, use of vitamin-mineral supplements, food supplement/herbal products, different dietary patterns like ketogenic diet or intermittent fasting, elimination diets, and eating disorders/food craving. The questions and answers in both languages are presented in Supplementary Table 1.

The questions were asked to the chatbot in both English and Turkish between May and June 2024 by the author, who was not involved in evaluating the answers provided by ChatGPT-4. Additionally, questions in both languages were asked again on a different day using the “new chat” button to evaluate the reproducibility of ChatGPT-4. Question and answer files were made for the English and Turkish versions, including the initial answers and the second answers acquired by using the “new chat” button.

Because ChatGPT-4 used in the present study is a public application, and there is no human/animal participant, ethics committee approval was not required.

2.2. Assessment of reliability, quality, and readability of responses

Reliability, quality, and readability were evaluated in both languages. Subjective scales such as DISCERN and global quality scale (GQS) were used assessing the reliability and quality. The evaluation of the answers was carried out by 12 independent experts, comprising 2 authors and a panel of 10 expert dietitians who have at least 10 years of experience as a dietitian; this expert panel was assembled due to the absence of established international recommendations specifically on PCOS and nutrition. The evaluation results were presented as the average of the scores given by the 12 independent experts.

The DISCERN is a 3-part scale based on subjective scoring used for the quality and reliability of health information [22]. The first part of the scale consists of 8 questions that assess the reliability of the text. The second part of the scale consists of 7 questions that evaluate the quality of information about treatment options and the last part of the scale includes a single question that evaluates the overall quality of the publication as an information source. However, because the reliability of ChatGPT's answers was being tested and no questions about the treatment were included, the DISCERN scale was modified (modified DISCERN [mDISCERN]) and the reliability of the responses was evaluated by using only the first part of the scale, similar to its use in other recent studies [23,24]. For each question on the mDISCERN scale, ChatGPT's answers were scored from 1 to 5, and the points received for each answer were summed. A total score below 40% (8–15 points) was categorized as poor, 40%–79% (16–31 points) as fair, and above 80% (32–40 points) as good [22].

Table 1 – Tools used to evaluate the readability of ChatGPT's responses both in English and Turkish

Tools	Formula	Evaluations	Ref.
Text in English			
FRE	$206.835 - 1015 \times (W/S) - 84.6 \times (B/W)$	The higher the score, the easier the text is to read.	[26]
GFI	$0.4 \times [(W/S) + 100 \times (C^*/W)]$	The GFI is used to estimate the educational level required to understand the text.	[27]
FKGL	$0.39 \times (W/S) + 11.8 \times (B/W) - 15.59$	The FKGL score represents the equivalent level of education in the United States.	[28]
CLI	$(0.0588 \times L) - (0.296 \times S^*) - 15.8$	CLI corresponds to the U.S. reading grade level.	[29]
SMOG	$1.0430 \times \sqrt{C} + 3.1291$	SMOG is used to determine the number of years of education an average person needs to understand a given piece of writing.	[30]
Text in Turkish			
Ateşman	$8.825 - 40.175 \times (B/W) - 2.610 \times (W/S)$	Adapted from FRE to Turkish. The higher the score, the easier the text is to read.	[31]
Bezirci-Yılmaz	$\sqrt{MW} \times (W3 \times 0.84) + (W4 \times 1.5) + (W5 \times 3.5) + (W6 \times 26.35)$	The formula is used to estimate the educational level required to understand the text in Turkey.	[32]
Çetinkaya-Uzun	$118,823 - (25,987 \times AWL) - (0,971 \times ASL)$	The formula provides information about reading level	[33]

Abbreviations: ASL, average sentence length; AWL, average word length; B, number of syllables; C, complex words (≥ 3 syllables); C*, complex words with exceptions including, proper nouns, words made 3 syllables by addition of “ed” or “es”, compound words made of simpler words; ChatGPT, Chat Generative Pretrained Transformer; CLI, Coleman–Liau Index; FKGL, Flesch–Kincaid Grade Level; FRES, Flesch Reading Ease Score; GFI, Gunning Fog Index; L, average number of letters per 100 words; MW, mean words in a sentence; S, number of sentences; S*, average number of sentences per 100 words; SMOG, Simple Measure of Gobbledygook; W, number of words; W3, number of 3-syllable words; W4, number of 4-syllable words; W5, number of 5-syllable words; W6, number of 6-syllable words.

The quality of ChatGPT's responses were evaluated using GQS. On this scale, each response was scored between 1 and 5 points: 1–2 points indicate low quality, 3 points indicate moderate quality, and 4–5 points indicate high quality [22].

The readability of the responses was objectively measured using commonly applied formulas [25]. Five validated and widely used readability evaluating tools were used for the English text: Flesch Reading Ease (FRE) score [26], Gunning Fog Index [27], Flesch–Kincaid Grade Level [28], Coleman–Liau Index [29], and Simple Measure of Gobbledygook tools [30]. Among these readability tools, an increase in the FRE score and a decrease in the other 4 scores indicate improved readability. The readability scores were computed using free online readability calculator tools [25]. For the Turkish text, readability was measured using 3 different formulas developed by Ateşman [31], Bezirci-Yılmaz [32], and Çetinkaya and Uzun [33]. The formulas used for calculations and the evaluation of the responses in both English and Turkish languages are shown in Table 1.

2.3. Statistical analysis

Agreement between 12 independent experts evaluating ChatGPT's answers was assessed using the Fleiss' Kappa coefficient. Numerical data were presented as mean $\pm$ standard deviation and categorical data were presented as numbers (percentages). The relationship between reliability, quality, and readability for both languages were assessed using Spearman rank correlation test because of the nonparametric data. Furthermore, the associations between readability indexes, mDISCERN, and GQS scores were analyzed using linear regression. Statistical analyses were performed using SPSS version

23 (SPSS, Chicago, IL, USA), with the significance level set at $P < .05$.

3. Results

The Fleiss' Kappa coefficients between 12 independent evaluators were 0.77 and 0.85 for reliability and 0.83 and 0.91 for quality in the first and second answers obtained from ChatGPT, respectively. These results indicate a substantial or almost perfect agreement between the 12 dietitians' ratings. Additionally, ChatGPT's reproducibility was 100% in both English and Turkish. When assessed by evaluator dietitians, it was found that no misleading information was provided.

3.1. Reliability and quality

The results obtained from the mDISCERN and GQS regarding the reliability and quality of the responses are shown in Table 2. The mean mDISCERN score of responses to 11 questions given by 12 independent dietitians was 27.6 ± 0.87 in English and 27.2 ± 0.87 in Turkish. When classified according to the mDISCERN score, all responses were in the “fair” (40%–79%; 16–31 points) category in both languages (Table 2).

The mean GQS score given by 12 independent dietitians was 4.5 ± 0.31 in English and 4.2 ± 0.43 in Turkish, indicating that the responses were of high quality based on the mean GQS score in both languages. However, when the scores assigned to the answers were evaluated by grouping them according to their quality, it was found that 9.1% of the responses in Turkish were of “moderate quality” (3 points) and 90.9% were of “high quality” (4–5 points), whereas all of the responses in English received scores in the “high quality” (4–5 points) category (Table 2).

Table 2 – Results regarding mDISCERN and GQS scores of the ChatGPT's responses to questions from women with PCOS

	Score	Evaluation
Text in English		
mDISCERN	27.6±0.87	Fair
Fair	100.0%	
GQS	4.5±0.31	High quality
High quality	100.0%	
Text in Turkish		
mDISCERN	27.2±0.87	Fair
Fair	100.0%	
GQS	4.2±0.43	High quality
Moderate quality	9.1%	
High quality	90.9%	

Abbreviations: ChatGPT, Chat Generative Pretrained Transformer; GQS, global quality scale; mDISCERN, modified DISCERN; PCOS, polycystic ovary syndrome.

Table 3 – Results regarding readability of the ChatGPT's responses to questions from women with PCOS

	Score	Reading level
Text in English		
FRE	26.86±8.10	Very difficult to read
GFI	18.93±1.71	Extremely difficult to read
FKGL	14.96±1.49	College graduate
CLI	15.80±1.12	College graduate
SMOG	13.65±1.16	College level
Text in Turkish		
Ateşman	48.7±8.96	Difficult to read
Bezirci-Yilmaz	19.7±5.28	Academic level
Çetinkaya-Uzun	30.8±4.37	College level

ChatGPT, Chat Generative Pretrained Transformer; CLI, Coleman-Liau Index; FKGL, Flesch-Kincaid Grade Level; FRES, Flesch Reading Ease Score; GFI, Gunning Fog Index; PCOS, polycystic ovary syndrome; SMOG, Simple Measure of Gobbledygook.

3.2. Readability

Results on the readability of responses in both languages are shown in Table 3. When the readability of English responses was evaluated, it was found that the readability levels ranged from very difficult to extremely difficult according to FRE and the Gunning Fog Index, with the readers level falling in the college to college graduate category according to the Flesch-Kincaid Grade Level, CLI, and Simple Measure of Gobbledygook measures. Moreover, the readability levels of the Turkish responses were categorized as difficult according to Ateşman formula, whereas readers level was classified at academic to college level according to Bezirci-Yılmaz and Çetinkaya-Uzun formulas (Table 3).

3.3. Relationship between reliability, quality, and readability

Spearman correlation test was applied to examine the correlations between reliability, quality, and readability for the Turkish and English versions; the results are presented in Table 4. A positive correlation was found between mDISCERN and GQS ($r = 0.449$, $P = .036$), whereas no correlation was observed between mDISCERN and GQS and readability indexes for the English text ($P > .05$). For the Turkish text, there was a positive correlation between mDISCERN and GQS ($r = 0.519$, $P = .013$), between GQS and Ateşman readability index ($r = 0.629$, $P = .002$), and between GQS and Çetinkaya-Uzun readability index ($r = 0.564$, $P = .006$). However, there is a negative correlation between GQS and Bezirci-Yılmaz readability index ($r = -0.440$, $P = .041$). Additionally, no correlation was found between mDISCERN and the readability indexes for Turkish text ($P > .05$) (Table 4).

Further, a linear regression model was applied to evaluate the association between readability indexes and quality and reliability scores; the results are presented in Table 5. The analysis revealed that the English readability indexes showed no significant association with either the DISCERN or GQS scores ($P > .05$). On the other hand, the Turkish readability indexes showed no significant association with the mDISCERN score, whereas the GQS score had a positive association with the Ateşman ($\beta = 0.570$, $P = .006$) and Çetinkaya-Uzun ($\beta = 0.571$, $P = .006$) readability indexes and a negative association with the Bezirci-Yılmaz index ($\beta = -0.490$, $P = .021$) (Table 5).

4. Discussion

ChatGPT is an advanced AI technology that can quickly collect and analyze data on a variety of subjects [8,11,34]. The results obtained from this study indicate that ChatGPT's responses are at an acceptable level of reliability and do not contain any misleading information when providing general nutritional advice to women with PCOS. To the best of our knowledge, this study is the first in the literature investigating the effectiveness of ChatGPT's responses on nutrition in PCOS.

Recent PCOS guideline include four dietary recommendations, which are the same as those for the general population. There is no separate nutrition guideline specifically focusing on PCOS [2]. The guideline suggest that no single dietary approach or component is superior to others. Personalized nutrition plans that avoid overly restrictive or unbalanced recommendations may increase dietary compliance, and a sustainable nutrition plan should be implemented [2]. Because the dietary questions frequently asked by women with PCOS and evaluated in this study (e.g., regarding different dietary patterns, eliminations, and supplements) go beyond general dietary recommendations, the answers provided by ChatGPT could not be directly evaluated against established guidelines. In this context, international studies have found that the general population's dietary habits are often far from the nutritional recommendations published for their respective communities, showing a tendency toward more insufficient and unbalanced diets [35,36]. However, to our knowledge, no such studies have been conducted in Turkey. Given that the general

Table 4 – The relationship between readability indexes, mDISCERN, and GQS scores of the ChatGPT's responses to questions from women with PCOS for both languages

	Text in English		Text in Turkish	
	mDISCERN	GQS	mDISCERN	GQS
FRE	0.080 (0.722)	−0.068 (0.763)	-	-
GFI	−0.227 (0.310)	0.015 (0.948)	-	-
FKGL	−0.193 (0.390)	−0.132 (0.560)	-	-
CLI	−0.023 (0.918)	0.146 (0.516)	-	-
SMOG	−0.237 (0.288)	0.010 (0.965)	-	-
Ateşman	-	-	0.216 (0.334)	0.629 (0.002)
Bezirci-Yilmaz	-	-	−0.115 (0.610)	−0.440 (0.041)
Çetinkaya-Uzun	-	-	0.191 (0.393)	0.564 (0.006)
mDISCERN	1	0.449 (0.036)	1	0.519 (0.013)
GQS	0.449 (0.036)	1	0.519 (0.013)	1

Spearman's rank correlation test was performed.

Abbreviations: ChatGPT, Chat Generative Pretrained Transformer; CLI, Coleman–Liau Index; FKGL, Flesch–Kincaid Grade Level; FRES, Flesch Reading Ease Score; GFI, Gunning Fog Index; GQS, global quality score; mDISCERN, modified DISCERN; PCOS, polycystic ovary syndrome; SMOG, Simple Measure of Gobbledygook.

Table 5 – The association between readability indexes, mDISCERN, and GQS scores of the ChatGPT's responses to questions from women with PCOS

	Unstandardized coefficients			Standardized coefficients		Adj. R ²	F	P	DW
	B	SEs	CI (95%)	β	t				
Text in English									
mDISCERN									
FRE	−0.005	0.024	−0.055 to 0.045	−0.047	−0.210	−0.048	0.044	.836	1.204
GFI	−0.047	0.113	−0.282 to 0.188	−0.093	−0.418	−0.041	0.175	.680	1.199
FKGL	−0.066	0.129	−0.335 to 0.203	−0.113	−0.510	−0.037	0.260	.616	1.203
CLI	0.034	0.173	−0.327 to 0.395	0.044	0.196	−0.048	0.038	.847	1.189
SMOG	−0.060	0.166	−0.406 to 0.286	−0.081	−0.361	−0.043	0.131	.722	1.207
GQS									
FRE	0.014	0.008	−0.017 to 0.023	0.007	0.031	−0.050	0.001	.975	1.793
GFI	−0.016	0.040	−0.399 to 0.694	−0.089	−0.399	−0.042	0.159	.694	1.819
FKGL	−0.035	0.045	−0.129 to 0.059	−0.170	−0.770	−0.020	0.593	.450	1.805
CLI	0.024	0.061	−0.103 to 0.151	0.087	0.388	−0.042	0.151	.702	1.747
SMOG	−0.028	0.058	−0.475 to 0.640	−0.106	−0.475	−0.038	0.226	.640	1.830
Text in Turkish									
mDISCERN									
Ateşman	0.029	0.021	−0.014 to 0.072	0.303	1.422	0.046	2.021	.171	1.993
Bezirci-Yılmaz	−0.047	0.035	−0.121 to 0.026	−0.287	−1.342	0.037	1.800	.195	2.015
Çetinkaya-Uzun	0.067	0.042	−0.020 to 0.154	0.338	1.607	0.070	2.584	.124	1.971
GQS									
Ateşman	0.027	0.009	0.009–0.046	0.570	3.106	0.292	9.646	.006	1.403
Bezirci-Yılmaz	−0.040	0.016	−0.073 to −0.007	−0.490	−2.515	0.202	6.327	.021	1.261
Çetinkaya-Uzun	0.056	0.018	0.018–0.094	0.571	3.110	0.294	9.671	.006	1.409

Linear regression analysis was performed.

Abbreviations: ChatGPT, Chat Generative Pretrained Transformer; CLI, Coleman–Liau Index; FKGL, Flesch–Kincaid Grade Level; FRES, Flesch Reading Ease Score; GFI, Gunning Fog Index; GQS, global quality score; mDISCERN, modified DISCERN; PCOS, polycystic ovary syndrome; SMOG, Simple Measure of Gobbledygook.

population is believed to consume diets that fall significantly short of recommended guidelines, it is plausible that women with PCOS may exhibit similar dietary patterns.

Professional dietitians reported that the nutritional recommendations provided by ChatGPT in response to frequently asked nutritional questions were largely appropriate in stud-

ies evaluating the efficacy of the chatbots [18,17]. ChatGPT's responses to patients' questions about their medical conditions have also been examined in the literature along with its nutritional recommendations. In this context, it was recently reported that ChatGPT is a usable resource for patient education and for answering patients' questions in various

medical situations, including PCOS [13] and its related infertility [14], as well as orthopedic diseases [37], sports surgery-related diseases [38], amyloidosis [39], urinary tract infections [40], appendicitis [41], and hypothyroidism during pregnancy [24]. In line with the findings of previous studies on different medical disorders, ChatGPT's responses related to nutrition in PCOS were found to be reasonably reliable and of high quality in both languages in the current study (Table 2). Conversely, in a study evaluating simulated patient information for an orthopedic and trauma surgery procedure using ChatGPT, the mDISCERN scores of ChatGPT's responses were 2.4 of 5. Additionally, when assessing the usability of the answers, none were classified as "very useful" and 20% of the responses contained potentially harmful information [23]. Recent studies have shown that although ChatGPT's responses are generally considered reliable and accurate, there is still need for further improvement in both areas [23,24,41,42]. Although ChatGPT can be a useful for delivering general information and aiding healthcare professionals in patient education because of potential benefits like quick access to general information [17,33–36], further development is needed, especially in providing personalized treatment recommendations in complex conditions such as PCOS and its integration in clinical practice must be carefully evaluated. In this regard, The Academy of Nutrition and Dietetics has reported that ChatGPT and other AI chatbots can assist registered dietitian nutritionists and nutrition dietetics technicians but cannot substitute the expertise, judgment, and social skills of health care professionals [12,16].

The reliability and quality of the information provided by AI chatbots such as ChatGPT are crucial. In addition, the readability and reading level of the responses are also extremely important [43,44]. In this context, the American Medical Association (AMA) [45] and National Institutes of Health (NIH) [46] recommend that written materials ideally be at sixth- and eighth-grade reading level, respectively, so that they are readable by the majority of adults for effective communication. In our study, the responses given by ChatGPT were classified as "difficult or very difficult to read," with text was written at college level and above both in English and Turkish (Table 3). Therefore, the readability level of the responses in the present study are not compatible with the recommendations made by the AMA and NIH [39,40]. Similarly, recent studies showed that readability of ChatGPT responses far exceeds the recommendations set by the AMA and NIH [39,41]. ChatGPT's responses may be challenging to understand, especially for individuals with lower health literacy levels. Therefore, guidance from health care providers is needed when interpreting such responses from the current version of ChatGPT for practical use and patient education.

Readability indexes were not found to be significantly related to reliability and accuracy of responses in English, whereas a significant relationship was found between reliability and accuracy of responses in Turkish in the present study (Table 4 and Table 5). Research has emphasized that simplifying language and using clear, plain language can improve patient understanding and health outcomes [47,48]. At this point, improving the readability of ChatGPT responses without compromising their reliability and quality is important. Consequently, even if ChatGPT's answers are plausibly reliable

and accurate, difficulty in reading them would prevent them from being easily understood by non-health care professionals and patients. Therefore, to improve readability without affecting reliability and accuracy, removing medical terms and creating easier-to-read text should be considered for future improvements.

Our study has some limitations. First, validated tools need to be developed to evaluate ChatGPT responses, which would eliminate heterogeneity in comparing the obtained data with the literature. Second, although there is consistency between the first and second answers received by ChatGPT and the reliability of these answers is high, there is no single correct answer to a question. Finally, ChatGPT-4 contains information only up to 2021; and no more recent information is included.

5. Conclusion

The rise of AI applications and the tendency of patients to consult these platforms necessitate that the information provided by these platforms be reviewed by experts in the field. In the present study, ChatGPT exhibited a methodical approach to answering questions and advised consulting health care professionals for personalized advice tailored to the specific health needs and circumstances of patients. Although ChatGPT's responses were generally reliable and of high quality, there is still room for improvement to enhance reliability and quality. ChatGPT should provide information at a more accessible reading level so that it can be understood by the majority of individuals especially with those lower health literacies in the society. Although ChatGPT offers reasonably accurate general nutritional recommendations for PCOS, it cannot replace personalized medical advice from qualified health care professionals. The integration of ChatGPT in clinical practice requires careful consideration, especially considering the complexity of this syndrome. For this reason, support from a health care professional is required during the use of ChatGPT in patient education and clinical practice. This study lays the groundwork for future research that could focus on improving AI-based nutrition recommendations with a particular focus on real-world applicability.

Author Declarations

The authors declare no competing interest.

CRedit authorship contribution statement

Elif Ulug: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Irmak Gunesli:** Writing – original draft, Data curation, Conceptualization. **Aylin Acikgoz Pinar:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Bulent Okan Yildiz:** Writing – review & editing, Project administration, Methodology, Conceptualization.

Acknowledgments

Authors have no acknowledgments to declare.

Source of Support

This research received no specific grant from any funding agency, commercial, or not-for-profit sectors.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.nutres.2024.11.005](https://doi.org/10.1016/j.nutres.2024.11.005).

REFERENCES

- [1] Azziz R, Carmina E, Chen Z, Dunaif A, Laven JS, Legro RS, et al. Polycystic ovary syndrome. *Nat Rev Dis Primers*. 2016;2:16057. doi:[10.1038/nrdp.2016.571](https://doi.org/10.1038/nrdp.2016.571).
- [2] Teede HJ, Tay CT, Laven JJE, Dokras A, Moran LJ, Piltonen TT, et al. Recommendations from the 2023 international evidence-based guideline for the assessment and management of polycystic ovary syndrome. *Eur J Endocrinol* 2023;189:G43–64. doi:[10.1093/eurjendo/evad096](https://doi.org/10.1093/eurjendo/evad096).
- [3] Stefanaki K, Karagiannakis DS, Peppas M, Vryonidou A, Kalantaridou S, Goulis DG, et al. Food cravings and obesity in women with polycystic ovary syndrome: pathophysiological and therapeutic considerations. *Nutrients* 2024;16. doi:[10.3390/nu16071049](https://doi.org/10.3390/nu16071049).
- [4] Hesse BW, Nelson DE, Kreps GL, Croyle RT, Arora NK, Rimer BK, et al. Trust and sources of health information: the impact of the internet and its implications for health care providers: findings from the first Health Information National Trends Survey. *Arch Intern Med* 2005;165:2618–24. doi:[10.1001/archinte.165.22.2618](https://doi.org/10.1001/archinte.165.22.2618).
- [5] Ouyang L, Wu J, Jiang X, Almeida D, Wainwright C, Mishkin P, et al. Training language models to follow instructions with human feedback. *Adv Neural Inf Process Syst* 2022;35:27730–44. doi:[10.48550/arXiv.2203.02155](https://doi.org/10.48550/arXiv.2203.02155).
- [6] Meskó B, Topol EJ. The imperative for regulatory oversight of large language models (or generative AI) in healthcare. *NPJ Digit Med* 2023;6:120. doi:[10.1038/s41746-023-00873-0](https://doi.org/10.1038/s41746-023-00873-0).
- [7] Thirunavukarasu AJ, Ting DSI, Elangovan K, Gutierrez L, Tan TF, Ting DSW. Large language models in medicine. *Nat Med* 2023;29:1930–40. doi:[10.1038/s41591-023-02448-8](https://doi.org/10.1038/s41591-023-02448-8).
- [8] Introducing ChatGPT. OpenAI, <https://openai.com/blog/chatgpt>. [accessed: 6 Oct 2024]
- [9] Bach RL, Wenz A. Studying health-related internet and mobile device use using web logs and smartphone records. *PLoS One* 2020;15:e0234663. doi:[10.1371/journal.pone.0234663](https://doi.org/10.1371/journal.pone.0234663).
- [10] Bujnowska-Fedak MM, Waligóra J, Mastalerz-Migas A. The internet as a source of health information and services, in advancements and innovations in health sciences, M. Pokorski, Editor. 2019, Springer International Publishing: Cham. 1–16.
- [11] Dave M, Patel N. Artificial intelligence in healthcare and education. *Br Dent J* 2023;234:761–4. doi:[10.1038/s41415-023-5845-2](https://doi.org/10.1038/s41415-023-5845-2).
- [12] Chatelan A, Clerc A, Fonta P-A. ChatGPT and future artificial intelligence chatbots: what may be the influence on credentialed nutrition and dietetics practitioners? *J Acad Nutr Diet* 2023;123:1525–31. doi:[10.1016/j.jand.2023.08.001](https://doi.org/10.1016/j.jand.2023.08.001).
- [13] Pace L, Kummer N, Bril F, Hosseinzadeh P, Azziz R. Evaluating ChatGPT's responses to frequently asked questions regarding polycystic ovary syndrome. *Fertil Steril* 2024;122:e55. doi:[10.1016/j.fertnstert.2024.05.100](https://doi.org/10.1016/j.fertnstert.2024.05.100).
- [14] Devranoglu B, Gurbuz T, Gokmen O. ChatGPT's efficacy in queries regarding polycystic ovary syndrome and treatment strategies for women experiencing infertility. *Diagnostics* 2024;14:1082. doi:[10.3390/diagnostics14111082](https://doi.org/10.3390/diagnostics14111082).
- [15] Arslan S. Decoding dietary myths: the role of ChatGPT in modern nutrition. *Clin Nutr ESPEN* 2024;60:285–8. doi:[10.1016/j.clnesp.2024.02.022](https://doi.org/10.1016/j.clnesp.2024.02.022).
- [16] Ponzo V, Goitre I, Favaro E, Merlo FD, Mancino MV, Riso S, et al. Is ChatGPT an effective tool for providing dietary advice? *Nutrients* 2024;16. doi:[10.3390/nu16040469](https://doi.org/10.3390/nu16040469).
- [17] Kirk D, van Eijnatten E, Camps G. Comparison of answers between ChatGPT and human dietitians to common nutrition questions. *J Nutr Metab* 2023;2023:5548684. doi:[10.1155/2023/5548684](https://doi.org/10.1155/2023/5548684).
- [18] Sun H, Zhang K, Lan W, Gu Q, Jiang G, Yang X, et al. An AI dietitian for type 2 diabetes mellitus management based on large language and image recognition models: preclinical concept validation study. *J Med Internet Res* 2023;25:e51300. doi:[10.2196/51300](https://doi.org/10.2196/51300).
- [19] Niszczota P, Rybicka I. The credibility of dietary advice formulated by ChatGPT: robo-diets for people with food allergies. *Nutrition* 2023;112:112076. doi:[10.1016/j.nut.2023.112076](https://doi.org/10.1016/j.nut.2023.112076).
- [20] Cetik S, Acikgoz A, Yildiz BO. Investigation of taste function and eating behavior in women with polycystic ovary syndrome. *Appetite* 2022;168:105776. doi:[10.1016/j.appet.2021.105776](https://doi.org/10.1016/j.appet.2021.105776).
- [21] Aksun S, Sonu NC, Aygun S, Karakulak UN, Mumusoglu S, Yildiz BO. Alterations of cardiometabolic risk profile in polycystic ovary syndrome: 13 years follow-up in an unselected population. *J Endocrinol Invest* 2024;47:1129–37. doi:[10.1007/s40618-023-02230-0](https://doi.org/10.1007/s40618-023-02230-0).
- [22] Kumar VS, Subramani S, Veerapan S, Khan SA. Evaluation of online health information on clubfoot using the DISCERN tool. *J Pediatr Orthop B* 2014;23:135–8. doi:[10.1097/bpb.0000000000000000](https://doi.org/10.1097/bpb.0000000000000000).
- [23] Kasapovic A, Ali T, Babasiz M, Bojko J, Gathen M, Kaczmarczyk R, et al. Does the information quality of chatgpt meet the requirements of orthopedics and trauma surgery? *Cureus* 2024;16:e60318. doi:[10.7759/cureus.60318](https://doi.org/10.7759/cureus.60318).
- [24] Onder CE, Koc G, Gokbulut P, Taskaldiran I, Kuskonmaz SM. Evaluation of the reliability and readability of ChatGPT-4 responses regarding hypothyroidism during pregnancy. *Sci Rep* 2024;14:243. doi:[10.1038/s41598-023-50884-w](https://doi.org/10.1038/s41598-023-50884-w).
- [25] Simpson D. The Readability Test Tool 2013, <http://www.readable.com>. [accessed: 6 Oct 2024]
- [26] Flesch R. A new readability yardstick. *J Appl Psychol* 1948;32:221–33. doi:[10.1037/h0057532](https://doi.org/10.1037/h0057532).
- [27] Gunning R. *The technique of clear writing*. New York: McGraw-Hill; 1968. Revised ed.
- [28] Kincaid JP, Fishburne RP Jr, Rogers RL, Chissom BS. Derivation of new readability formulas (automated readability index, fog count and Flesch reading ease formula) for navy enlisted personnel. *Res Branch Rep*. 1975;8–75.
- [29] Coleman M, Liau TL. A computer readability formula designed for machine scoring. *J Appl Psychol* 1975;60:283–4. doi:[10.1037/h0076540](https://doi.org/10.1037/h0076540).
- [30] Hedman AS. Using the SMOG formula to revise a health-related document. *Am J Health Educ* 2008;39:61–4. doi:[10.1080/19325037.2008.10599016](https://doi.org/10.1080/19325037.2008.10599016).
- [31] Ateşman E. Measuring readability in Turkish. *Dil Dergisi* 1997;58:71–4.

- [32] Bezirci B, Yilmaz AE. A software library for measurement of readability of texts and a new readability metric for Turkish. *Dokuz Eylül University Faculty of Engineering Journal of Science and Engineering* 2010;12(3):49–62.
- [33] Çetinkaya G. Defining and classifying the readability levels of Turkish texts. Ankara University Social Sciences Institute. Doctoral thesis, Ankara.
- [34] Ram B, Verma P. Artificial intelligence AI-based Chatbot study of ChatGPT, Google AI Bard and Baidu AI. *World J Adv Eng Technol Sci* 2023;8:258–61. doi:10.30574/wjaets.2023.8.1.0045.
- [35] Germain L, Lata arche C, Kesse-Guyot E, Galan P, Hercberg S, Briançon S. Does compliance with nutrition guidelines lead to healthy aging? A quality-of-life approach. *J Acad Nutr Diet* 2013;113:228–40 e1–2. doi:10.1016/j.jand.2012.10.015.
- [36] Malon A, Deschamps V, Salanave B, Vernay M, Szego E, Estaquio C, et al. Compliance with French nutrition and health program recommendations is strongly associated with socioeconomic characteristics in the general adult population. *J Am Diet Assoc* 2010;110:848–56. doi:10.1016/j.jada.2010.03.027.
- [37] Giorgino R, Alessandri-Bonetti M, Del Re M, Verdoni F, Peretti GM, Mangiavini L. Google Bard and ChatGPT in orthopedics: which is the better doctor in sports medicine and pediatric orthopedics? The role of AI in patient education. *Diagnostics (Basel)* 2024;14. doi:10.3390/diagnostics14121253.
- [38] Yüce A, Erkurt N, Yerli M, Misir A. The potential of ChatGPT for high-quality information in patient education for sports surgery. *Cureus* 2024;16:e58874. doi:10.7759/cureus.58874.
- [39] King RC, Samaan JS, Yeo YH, Peng Y, Kunkel DC, Habib AA, et al. A multidisciplinary assessment of ChatGPT's knowledge of amyloidosis: observational study. *JMIR Cardiol* 2024;8:e53421. doi:10.2196/53421.
- [40] Cakir H, Caglar U, Sekkeli S, Zerdali E, Sarilar O, Yildiz O, et al. Evaluating ChatGPT ability to answer urinary tract infection-related questions. *Infect Dis Now* 2024;54:104884. doi:10.1016/j.idnow.2024.104884.
- [41] Ghanem YK, Rouhi AD, Al-Houssan A, Saleh Z, Moccia MC, Joshi H, et al. Dr. Google to Dr. ChatGPT: assessing the content and quality of artificial intelligence-generated medical information on appendicitis. *Surg Endosc* 2024;38:2887–93. doi:10.1007/s00464-024-10739-5.
- [42] Barlas T, Altinova AE, Akturk M, Toruner FB. Credibility of ChatGPT in the assessment of obesity in type 2 diabetes according to the guidelines. *Int J Obes (Lond)* 2024;48:271–5. doi:10.1038/s41366-023-01410-5.
- [43] Seurer AC, Vogt HB. Low health literacy: a barrier to effective patient care. *South Dakota Med* 2013;66(2):51–57.
- [44] Wynia MK, Osborn CY. Health literacy and communication quality in health care organizations. *J Health Commun* 2010;15(2):102–15 Suppl. doi:10.1080/10810730.2010.499981.
- [45] Weiss BD. Health literacy: a manual for clinicians. *American Medical Association* 2003:253–358.
- [46] Easy-to-read health information. MedlinePlus, https://medlineplus.gov/all_easytoread.html. [Accessed: Oct 6 2024].
- [47] Kreps GL. Promoting patient comprehension of relevant health information. *Isr J Health Policy Res* 2018;7:56. doi:10.1186/s13584-018-0250-z.
- [48] Sørensen K, Van den Broucke S, Fullam J, Doyle G, Pelikan J, Slonska Z, et al. Health literacy and public health: a systematic review and integration of definitions and models. *BMC Public Health* 2012;12:80. doi:10.1186/1471-2458-12-80.